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COMPUTER NETWORKS (CSA0711)

20 MULTIPLE CHOICE QUESTIONS – ANSWERS & JUSTIFICATIONS

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

a) Slow Start

b) Fast Retransmit

c) Fast Recovery

d) Tahoe Reset

Answer: c) Fast Recovery

Why: Fast Retransmit triggers on 3 duplicate ACKs, but Fast Recovery is the mechanism that halves the congestion window.

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:

a) 9 MSS

b) 16 MSS

c) 12 MSS

d) 8 MSS

Answer: b) 16 MSS

Why: Since CWND (8) < SSTHRESH (16), the sender is in slow start, where CWND doubles each RTT with no loss: 8 → 16.

3. Which field in the TCP header is used to implement flow control at the receiver side?

a) Sequence number

b) Acknowledgement number

c) Window size

d) Urgent pointer

Answer: c) Window size

Why: The receiver advertises available buffer space via the Window Size field.

4. Nagle's algorithm causes increased latency primarily because it:

a) Disables ACK piggybacking

b) Buffers small segments until ACK arrives

c) Reduces CWND on every packet

d) Increases SYN backlog

Answer: b) Buffers small segments until ACK arrives

Why: Nagle's algorithm holds small segments until a pending ACK is received before sending, adding delay.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

a) MTU size

b) CWND growth rate relative to RTT

c) DNS resolution time

d) UDP competition